
COMBINATORICS & ALGORITHMS

Matroid Theory

from First Principles & Its Role in Greedy Algorithms

A rigorous yet accessible introduction
for advanced undergraduates and early graduate students

Topics: Matroid Axioms · Greedy Theorem · Graphic Matroids · MST Algorithms

Part I

Motivation

Why do some greedy algorithms work — and others fail?

I

The Central Question: When Does Greedy Work?

When does greedily choosing the locally best option yield a globally optimal solution?

- Greedy algorithms are simple, fast, and elegant.
- But they are *not always correct*.
- **Matroids** reveal exactly when and why greedy is provably optimal.

Answer: The greedy algorithm finds a maximum-weight solution for *every* weight function **if and only if** the feasible sets form a matroid.

A Greedy Success: Select the k Largest Items

Problem: Choose at most k items from a set to maximise total value.

Example: Greedy Works!

Items with values $\{9, 5, 3, 1\}$, $k = 2$.

Greedy: pick 9, then 5. Total = 14. This is optimal.

Why does greedy work here? • Any subset of size $\leq k$ is feasible.

- If our set has fewer than k items and a better unused item exists, we can freely add it.
- No feasibility constraint ever *blocks* a greedy choice.
 - This is the **uniform matroid** in disguise — we will formalise it shortly.

A Greedy Failure: 0/1 Knapsack

Problem: Capacity = 10. Greedy by value/weight ratio:

Item	Weight	Value	Val/Wt
A	6	12	2.0
B	5	9	1.8
C	5	9	1.8

Non-Example: Greedy by ratio fails!

Greedy picks A (weight 6), then cannot fit B or C . Total value = 12.

Optimal: $B + C$ (weight = 10, value = 18).

- The feasible sets $\{S : \text{weight}(S) \leq 10\}$ do **not** form a matroid.
- The **exchange axiom** fails — this is why greedy breaks.

Independence: The Common Abstraction

Both examples involve a collection of *feasible subsets*. The key is their structure:

Setting	Ground Set E	Independent Sets $\mathcal{I}$
Choose $\leq k$ items	Items	Subsets of size $\leq k$
Linear algebra	Vectors	Linearly independent subsets
Graph theory	Edges	Acyclic edge sets (forests)
Scheduling	Jobs	Feasible job subsets

A **matroid** $M = (E, \mathcal{I})$ abstracts precisely the structure shared by all of these — enabling a single unified proof of greedy correctness.

Part II

First Principles

Defining matroids from axioms

II

Definition of a Matroid

Definition: Matroid $M = (E, \mathcal{I})$

A **matroid** is a pair $M = (E, \mathcal{I})$ where E is a finite set and $\mathcal{I} \subseteq 2^E$ satisfies:

(I1) **Non-emptiness:** $\emptyset \in \mathcal{I}$

(I2) **Heredity:** If $A \in \mathcal{I}$ and $B \subseteq A$, then $B \in \mathcal{I}$

(I3) **Exchange:** If $A, B \in \mathcal{I}$ and $|A| < |B|$, then $\exists x \in B \setminus A$ such that $A \cup \{x\} \in \mathcal{I}$

- Elements of $\mathcal{I}$ are called **independent**; others are **dependent**.
- E is the **ground set**. Examples: columns of a matrix, edges of a graph, items in a set.

Axiom	Name	Informal Meaning
(I1)	Non-emptiness	Choosing nothing is always feasible
(I2)	Heredity	Subsets of feasible sets are feasible
(I3)	Exchange	Smaller independent sets can be augmented

Axiom (I1): Non-Emptiness

(I1) The empty set is independent: $\emptyset \in \mathcal{I}$

Interpretation:

- Choosing nothing is always feasible — a base case for building independent sets.
- Without (I1), we could not guarantee any solution exists.

In each setting:

- **Vectors:** The empty set of vectors is trivially linearly independent.
- **Edges:** The empty edge set is a trivially acyclic forest.
- **Selection:** Choosing zero items satisfies any cardinality bound.

Axiom (I2): Heredity (Downward Closure)

(I2) If $A \in \mathcal{I}$ and $B \subseteq A$, then $B \in \mathcal{I}$

Interpretation:

Subsets of independent sets are independent. Independence is *downward-closed*.

In each setting:

- **Vectors:** Any subset of linearly independent vectors is linearly independent.
- **Edges:** Any subset of an acyclic edge set is also acyclic.
- **Selection:** If choosing k items is feasible, choosing fewer is also feasible.

Non-Example: Does the Knapsack feasible system violate heredity?

Feasible sets are $\{S : \text{weight}(S) \leq 10\}$. $\{B, C\}$ is feasible as well as $\{A\}$. The issue: adding B or C to $\{A\}$ *exceeds* capacity, so $\{A, B\} \notin \mathcal{I}$. The downward-closure property *does* hold here individually, but exchange fails.

Axiom (I3): Exchange (Augmentation) — The Key Axiom

(I3) If $A, B \in \mathcal{I}$ and $|A| < |B|$, then $\exists x \in B \setminus A$ such that $A \cup \{x\} \in \mathcal{I}$

Interpretation:

Any smaller independent set can be *augmented* from a larger one.

- We can always “fill up” a small independent set using elements of a bigger one.
- This is the **heart of greedy correctness** — we prove this formally in Part IV.
- In linear algebra: if $\dim(A) < \dim(B)$, we can find a vector in B not in $\text{span}(A)$.

Why this matters algorithmically:

- Exchange prevents greedy from reaching a “dead end” — a locally maximal but globally sub-optimal solution.
- It implies all maximal independent sets have equal size — no “accidentally small” solutions.

Basic Terminology: Bases and Rank

Definition: Basis

A **basis** of a matroid $M = (E, \mathcal{I})$ is a maximal independent set: $B \in \mathcal{I}$ and no element $e \in E \setminus B$ can be added while staying independent.

Definition: Rank

For $A \subseteq E$, the **rank** of A is the size of the largest independent subset of A :

$$r(A) = \max_{I \subseteq A, I \in \mathcal{I}} |I|.$$

The **rank of the matroid** is $r(E)$.

These terms are introduced here because they are used in the examples below. Part III develops them further, including circuits and closure.

Uniform Matroids $U_{k,n}$

Definition: Uniform Matroid $U_{k,n}$

Ground set: $E = \{1, 2, \dots, n\}$.

Independent sets: $\mathcal{I} = \{A \subseteq E : |A| \leq k\}$.

Example: $U_{2,4}$, $E = \{a, b, c, d\}$, $k = 2$

$\mathcal{I} = \{\emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}\}$

Dependent sets: $\{a, b, c\}, \{a, b, d\}, \{a, c, d\}, \{b, c, d\}, \{a, b, c, d\}$

Axiom verification:

- (I1) $\emptyset \in \mathcal{I}$ since $|\emptyset| = 0 \leq k$.
- (I2) If $|A| \leq k$ and $B \subseteq A$, then $|B| \leq |A| \leq k$.
- (I3) If $|A| < |B| \leq k$, pick any $x \in B \setminus A$. Then $|A \cup \{x\}| = |A| + 1 \leq |B| \leq k$.
- **Greedy on $U_{k,n}$:** simply select the k highest-weight elements. Trivially optimal.

Linear Matroids

Definition: Linear Matroid

Let $\mathbb{F}$ be a field, A an $m \times n$ matrix over $\mathbb{F}$.

Ground set: $E = \{1, \dots, n\}$ (column indices).

Independent sets: $\mathcal{I} = \{S \subseteq E : \text{columns indexed by } S \text{ are linearly independent}\}.$

Example: Over $\mathbb{R}$

$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$. Columns: $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $v_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Independent: $\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}$.

Dependent: $\{1, 2, 3\}$ since $v_3 = v_1 + v_2$.

- Exchange axiom $\leftrightarrow$ the linear-algebra augmentation lemma for bases.
- Rank of matroid = column rank of A .

Graphic Matroids $M(G)$

Definition: Graphic Matroid $M(G)$

Let $G = (V, E)$ be an undirected graph.

Ground set: E (edges of G).

Independent sets: $\mathcal{I} = \{F \subseteq E : (V, F) \text{ is a forest (acyclic)}\}$.

Structure:

- Bases = spanning forests; in connected G : bases = spanning trees (each with $|V| - 1$ edges).
- Rank: $r(F) = |V| - (\text{number of components of } (V, F))$.

Axiom verification:

- (I1) Empty edge set is an acyclic forest.
- (I2) Any subset of an acyclic edge set is acyclic.
- (I3) Two forests F_1, F_2 with $|F_1| < |F_2|$: F_2 has fewer components, so some edge of F_2 connects two components of F_1 — adding it stays acyclic.

A Non-Example: Failure of the Exchange Axiom

Non-Example: Non-matroid: mismatched maximal sets

$E = \{a, b, c\}$, $\mathcal{I} = \{\emptyset, \{a\}, \{b, c\}\}$.

Both $\{a\}$ and $\{b, c\}$ are maximal (cannot add any element).

But $|\{a\}| = 1 \neq 2 = |\{b, c\}|$.

Exchange fails: from $\{b, c\}$, we cannot augment $\{a\} - \{a, b\} \notin \mathcal{I}$ and $\{a, c\} \notin \mathcal{I}$.

Non-Example: Diagnostic

If (I3) fails, there exist weights making greedy suboptimal.

Proof: Assign $w(a) = 5$, $w(b) = w(c) = 3$.

Greedy picks $\{a\}$ (value 5). But $\{b, c\}$ has value 6. Greedy fails.

Key lesson: Always verify the matroid axioms before trusting a generic greedy algorithm. For a hereditary independence system, (I3) is the decisive axiom: greedy is optimal for every weight function if and only if exchange holds. If exchange fails, some weight function makes greedy fail.

Part III

Structure

Bases · Rank · Circuits · Closure

III

Bases Revisited

Definition: Basis (full statement)

A **basis** of $M = (E, \mathcal{I})$ is a maximal independent set: $B \in \mathcal{I}$ such that $B \cup \{e\} \notin \mathcal{I}$ for all $e \in E \setminus B$.

The collection of all bases is denoted $\mathcal{B}(M)$.

Theorem: Equal Cardinality of Bases

All bases of a matroid have the same cardinality.

Proof sketch: Suppose $|B_1| < |B_2|$. By (I3), $\exists e \in B_2 \setminus B_1$ with $B_1 \cup \{e\} \in \mathcal{I}$, contradicting maximality of B_1 . ■

Examples:

- $U_{k,n}$: all bases have size exactly k .
- Linear matroid on A : bases are maximal linearly independent column sets — all have size = $\text{rank}(A)$.
- Graphic matroid on connected G : bases = spanning trees, all of size $|V| - 1$.

Rank Function Revisited

Definition: Rank $r : 2^E \rightarrow \mathbb{Z}_{\geq 0}$ (full statement)

For $A \subseteq E$: $r(A) = \max\{|I| : I \in \mathcal{I}, I \subseteq A\}$.

The **rank of the matroid** is $r(E)$.

Key properties:

- (R1) $0 \leq r(A) \leq |A|$
- (R2) **Monotone**: $A \subseteq B \Rightarrow r(A) \leq r(B)$
- (R3) **Submodular**: $r(A \cup B) + r(A \cap B) \leq r(A) + r(B)$

Let $C = A \cap B$. Since $C \subseteq A$ and $B \subseteq A \cup B$, monotonicity gives

$$r(C) \leq r(A), \quad r(B) \leq r(A \cup B).$$

The rank gained by adding A to the larger set B is no more than the rank gained by adding A to C :

$$r(A \cup B) - r(B) \leq r(A) - r(C).$$

Using $C = A \cap B$ and rearranging,

$$r(A \cup B) + r(A \cap B) \leq r(A) + r(B).$$

In specific matroids:

- $U_{k,n}$: $r(A) = \min(|A|, k)$.
- Linear matroid: $r(A) = \text{column rank of the submatrix with columns in } A$.
- Graphic matroid: $r(F) = |V| - c(F)$ where $c(F) = \text{number of connected components}$.

Circuits

Definition: Circuit

A **circuit** is a minimal dependent set: $C \subseteq E$ with $C \notin \mathcal{I}$ but $C \setminus \{e\} \in \mathcal{I}$ for every $e \in C$.

The collection of all circuits is $\mathcal{C}(M)$.

In graphic matroids:

- Circuits = simple cycles of G .
- Adding any edge to a spanning tree creates exactly one circuit.
- Cycle property: the heaviest edge in any cycle belongs to no MST.

Circuit properties:

- No circuit is a subset of another.
- **Circuit elimination:** If $C_1, C_2 \in \mathcal{C}(M)$ and $e \in C_1 \cap C_2$, then $\exists C_3 \in \mathcal{C}(M)$ with $C_3 \subseteq (C_1 \cup C_2) \setminus \{e\}$.

Let $X = (C_1 \cup C_2) \setminus \{e\}$. Suppose no circuit lies in X . Then X is independent, so $r(X) = |X|$.

Since C_1, C_2 are circuits, $r(C_i) = |C_i| - 1$. By submodularity, $r(C_1 \cup C_2) + r(C_1 \cap C_2) \leq |C_1| + |C_2| - 2$.

But X independent gives $r(C_1 \cup C_2) = r(X) = |C_1 \cup C_2| - 1$,

and always

$$r(C_1 \cap C_2) \leq |C_1 \cap C_2|.$$

Hence

$$|C_1 \cup C_2| - 1 + |C_1 \cap C_2| \leq |C_1| + |C_2| - 2.$$

So

$$|C_1 \cup C_2| + |C_1 \cap C_2| \leq |C_1| + |C_2| - 1,$$

contradicting

$$|C_1 \cup C_2| + |C_1 \cap C_2| = |C_1| + |C_2|.$$

Therefore X is dependent and contains a circuit $C_3 \subseteq X$.

Closure (Span)

Definition: Closure $\text{cl} : 2^E \rightarrow 2^E$

$$\text{cl}(A) = \{e \in E : r(A \cup \{e\}) = r(A)\}.$$

Equivalently: $e \in \text{cl}(A)$ iff adding e to A does not increase rank.

Intuition:

- $\text{cl}(A)$ is the “span” of A – all elements already “dependent” on A .
- Linear matroids: $\text{cl}(A) = \{\text{vectors in } E \text{ lying in } \text{span}(A)\}$.
- Graphic matroids: $\text{cl}(F) = F \cup \{\text{edges whose endpoints are connected in } (V, F)\}$.

Closure axioms (alternative axiom system for matroids):

- (CL1) $A \subseteq \text{cl}(A)$
- (CL2) $A \subseteq B \Rightarrow \text{cl}(A) \subseteq \text{cl}(B)$
- (CL3) $\text{cl}(\text{cl}(A)) = \text{cl}(A)$
- (CL4) **Mac Lane–Steinitz:** $e \notin \text{cl}(A)$ and $e \in \text{cl}(A \cup \{f\}) \Rightarrow f \in \text{cl}(A \cup \{e\})$

Part IV

Greedy Algorithms & Matroids

The central theorem and its proof

IV

The Generic Greedy Algorithm for Matroids

```
GREEDY( $M = (E, \mathcal{I}), w : E \rightarrow \mathbb{R}$ ):  
Sort elements:  $e_1, e_2, \dots, e_n$  with  $w(e_1) \geq w(e_2) \geq \dots \geq w(e_n)$   
 $S \leftarrow \emptyset$   
for  $i \leftarrow 1 \dots n$  do: if  $S \cup \{e_i\} \in \mathcal{I}$ :  $S \leftarrow S \cup \{e_i\}$   
return  $S$ 
```

Properties:

- Time: $O(n \log n)$ for sorting + n independence oracle calls.
- Correctness: holds **if and only if** $M = (E, \mathcal{I})$ is a matroid.
- Independence oracle examples:
 - $U_{k,n}$: check $|S| \leq k$. $O(1)$.
 - Graphic: check cycle via Union-Find. $O(\alpha(n))$ amortised.
 - Linear: check rank via Gaussian elimination. $O(n^2)$.

The Matroid Greedy Theorem

Theorem: Matroid Greedy Theorem (Edmonds, 1971)

Let $(E, \mathcal{I})$ be an independence system (satisfying I1 and I2). The greedy algorithm finds a maximum-weight basis for **every** weight function $w : E \rightarrow \mathbb{R}_{\geq 0}$ **if and only if** $(E, \mathcal{I})$ is a matroid.

Proof sketch ($\Rightarrow$) – matroids make greedy optimal:

- Let greedy choose $S = \{s_1, \dots, s_k\}$ in decreasing weight order.
- We build an optimal basis O_i containing the greedy prefix $\{s_1, \dots, s_i\}$.
- Start with any optimal basis O_0 . Suppose O_{i-1} contains $G_{i-1} = \{s_1, \dots, s_{i-1}\}$.
- If $s_i \notin O_{i-1}$, then $O_{i-1} \cup \{s_i\}$ contains a circuit C with $s_i \in C$. Since $G_{i-1} \cup \{s_i\}$ is independent, C contains some $o \in O_{i-1} \setminus G_{i-1}$. Removing this o breaks the circuit, so

$$O_i = O_{i-1} \setminus \{o\} \cup \{s_i\}$$

is again a basis.

- Since $G_{i-1} \cup \{o\} \subseteq O_{i-1}$, the element o was feasible when greedy chose s_i . Hence

$$w(s_i) \geq w(o).$$

- So replacing o by s_i does not decrease total weight. Thus O_i is still optimal and contains the longer greedy prefix.
- After all steps, an optimal basis contains all of S ; since both are bases, it equals S . Therefore greedy is optimal. ■

Proof sketch ($\Leftarrow$) – failing (I3) breaks greedy:

- Construct weights that force greedy to pick a smaller maximal set over a larger one – demonstrating sub-optimality.

Exchange Axiom: Why Greedy Cannot Get Stuck

The exchange axiom is precisely what prevents greedy from reaching a “local maximum trap”.

Without exchange:

- Two maximal independent sets can have different sizes.
- Greedy might fill up a small maximal set, missing a larger (better-weight) one.
- No way to “repair” the partial solution.

With exchange:

- All maximal independent sets have equal size (= rank of matroid).
- Greedy’s partial solution is always extendable — it never gets permanently stuck.
- Any element missed by greedy could have been included (at the appropriate weight rank), contradicting greedy’s sorted order.

Theorem:

A family $\mathcal{I}$ satisfying (I1) and (I2) is a matroid $\Leftrightarrow$ all maximal independent sets have equal cardinality (for every restriction to a subset $A \subseteq E$).

Part V

Greedy in Action

Specialising to classical algorithms



Partition Matroid: Best From Each Category

Definition: Partition Matroid

E partitioned into disjoint classes $E_1, E_2, \dots, E_m$ with quotas $k_1, \dots, k_m$.

$\mathcal{J} = \{S \subseteq E : |S \cap E_i| \leq k_i \text{ for all } i\}$.

Example: Job scheduling by type

$E = \{\text{coding jobs}\} \cup \{\text{design jobs}\} \cup \{\text{testing jobs}\}$, quotas $k_1 = 2, k_2 = 1, k_3 = 2$.

Greedy: sort all jobs by value; pick each job if its category quota is not exceeded.

Greedy specialisation:

- Sort all elements by weight descending.
- Add element $e \in E_i$ if $|S \cap E_i| < k_i$.
- This is optimal by the matroid greedy theorem.

Application: Bipartite matching = common independent set of two partition matroids (matroid intersection).

Let $G = (L \cup R, E)$ be bipartite, and take the ground set to be the edges.

Define two partition matroids on E :

- M_L : at most one chosen edge incident to each $u \in L$.
- M_R : at most one chosen edge incident to each $v \in R$.

A set $F \subseteq E$ is independent in both iff no two edges in F share a left endpoint or a right endpoint. That is exactly a matching.

Hence bipartite matching is a **matroid intersection** problem:

$$F \in \mathcal{J}_L \cap \mathcal{J}_R.$$

Graphic Matroid: Maximum-Weight Forest

Matroid: $M(G) = (E, \mathcal{I})$ where $E = \text{edges}$, $\mathcal{I} = \text{acyclic subsets}$.

Generic greedy on $M(G)$:

- Sort edges by weight descending (for max forest) or ascending (for MST).
- Add each edge if it does not create a cycle (independence check via Union-Find).
- Output: a maximum-weight spanning forest.

Example: Small worked example

G : vertices $\{1, 2, 3, 4\}$, edges with weights: $e_{12} = 5, e_{23} = 3, e_{34} = 4, e_{14} = 2, e_{13} = 6$. Sorted desc: $e_{13} = 6, e_{12} = 5, e_{34} = 4, e_{23} = 3, e_{14} = 2$.

Step 1: Add e_{13} . Step 2: Add e_{12} . Step 3: Add e_{34} .

Step 4: e_{23} creates cycle $1 - 2 - 3 - 1$ — reject. Step 5: e_{14} creates cycle — reject.

Max forest: $\{e_{13}, e_{12}, e_{34}\}$, weight = 15.

For MST: negate weights (or sort ascending). Kruskal's algorithm is exactly this greedy!

Part VI

Minimum Spanning Trees

Kruskal · Prim · Cut and Cycle Properties

VI

The Minimum Spanning Tree Problem

Definition: Minimum Spanning Tree

Given: connected undirected graph $G = (V, E)$, weight $w : E \rightarrow \mathbb{R}$.

Find: a spanning tree $T \subseteq E$ minimising $\sum_{e \in T} w_e$.

Matroid connection:

- Graphic matroid $M(G) = (E, \mathcal{I})$: E = edges, $\mathcal{I}$ = forests.
- Bases = spanning trees. MST = minimum-weight basis of $M(G)$.
- By the matroid greedy theorem: processing edges in ascending weight order and adding cycle-free edges is optimal.

Applications:

- Network design: cable routing, road construction, pipeline layout.
- Clustering: single-linkage hierarchical clustering.
- Approximation algorithms: MST gives a 2-approximation for TSP.
- Phylogenetics, image segmentation, circuit layout.

Kruskal's Algorithm

```
KRUSKAL( $G = (V, E), w : E \rightarrow \mathbb{R}$ ):  
Sort edges:  $e_1, e_2, \dots, e_m$  with  $w(e_1) \leq w(e_2) \leq \dots \leq w(e_m)$   
 $T \leftarrow \emptyset$   
DSU.init( $V$ )  
for  $i \leftarrow 1 \dots m$ :  
    let  $e_i = (u, v)$   
    if DSU.find( $u$ )  $\neq$  DSU.find( $v$ ):  
         $T \leftarrow T \cup \{e_i\}$   
        DSU.union( $u, v$ )  
return  $T$ 
```

Analysis:

- Independence check: adding $e_i = (u, v)$ is independent $\Leftrightarrow u$ and v are in different components of (V, T) . DSU.find: $O(\alpha(|V|))$.
- Total time: $O(|E| \log |E|)$ dominated by sorting.
- Correctness: direct corollary of the matroid greedy theorem applied to $M(G)$.

Union-Find (Disjoint Set Union)

Purpose: Maintain a partition of V into connected components of the growing forest.

DSU.init(V): each vertex forms its own singleton component

DSU.find(u): return root/representative of the component containing u

DSU.union(u, v): merge the components of u and v

Key invariant: $\text{find}(u) = \text{find}(v) \Leftrightarrow u$ and v are connected in current T

Optimisations:

- **Union by rank:** always attach the shorter tree under the taller — keeps trees shallow.
- **Path compression:** during find, flatten all nodes directly to root — future finds faster.
- With both: amortised $O(\alpha(n))$ per operation, where α is the inverse Ackermann function.
- For all practical purposes: $\alpha(n) \leq 4$ for $n \leq 10^{80}$.

In Kruskal's: $|E|$ find operations + $|V| - 1$ union operations $\Rightarrow$ total DSU cost $\approx O(|E|)$.

Kruskal's Algorithm: Worked Example

Graph G : 5 vertices $\{A, B, C, D, E\}$, 7 edges.

Edge	Weight	Action
$C-E$	2	Add ✓
$A-B$	4	Add ✓
$D-E$	6	Add ✓
$B-E$	7	Add ✓
$A-D$	8	Skip (cycle)
$B-C$	8	Skip (cycle)
$B-D$	11	Skip (cycle)

MST found: $T = \{C-E, A-B, D-E, B-E\}$

Total weight = $2 + 4 + 6 + 7 = 19$

After adding $C-E$: components $\{C, E\}, \{A\}, \{B\}, \{D\}$.
 After $A-B$: $\{A, B\}, \{C, E\}, \{D\}$.
 After $D-E$: $\{C, D, E\}, \{A, B\}$.
 After $B-E$: $\{A, B, C, D, E\}$ — done!

Prim's Algorithm

```

PRIM( $G = (V, E), w : E \rightarrow \mathbb{R}, s \in V$ ):
  key[v]  $\leftarrow \infty$  for all  $v \in V$ 
  parent[v]  $\leftarrow \text{NIL}$  for all  $v \in V$ 
  key[s]  $\leftarrow 0$ 
   $Q \leftarrow$  min-priority queue containing all vertices, keyed by key[.]
  while  $Q \neq \emptyset$ :
     $u \leftarrow \text{EXTRACT-MIN}(Q)$ 
    for each neighbour  $v$  of  $u$ :
      if  $v \in Q$  and  $w(u, v) < \text{key}[v]$ :
        parent[v]  $\leftarrow u$ 
        key[v]  $\leftarrow w(u, v)$ 
        DECREASE-KEY( $Q, v, \text{key}[v]$ )
  return { (parent[v], v) :  $v \in V \setminus \{s\}$  }

```

Analysis:

- Assumes G is connected; otherwise this returns a spanning tree only for the component reachable from s .
- Time: $O(|E| \log |V|)$ with a binary heap; $O(|E| + |V| \log |V|)$ with a Fibonacci heap.
- Prim grows a **single tree** from s , repeatedly adding the cheapest edge from the current tree to a new vertex.
- Correctness: by the **cut property** — at each step, the cheapest edge crossing from the current tree to the remaining vertices is safe.

Cut Property and Cycle Property

Theorem: Cut Property

For any cut $(S, V \setminus S)$ of G , the minimum-weight edge crossing the cut belongs to some MST. If this edge is unique, it belongs to every MST.

Theorem: Cycle Property

For any cycle C in G , the maximum-weight edge of C does not belong to any MST. If this edge is unique, it belongs to no MST.

Connection to matroid structure:

- Cut property $\Leftrightarrow$ exchange axiom: a sub-optimal tree can be augmented.
- Cycle property $\Leftrightarrow$ circuit elimination: the heaviest circuit edge is excluded.

Algorithmic use:

- **Kruskal** uses the cycle property: skip edge if it creates a cycle.
- **Prim** uses the cut property: always take the cheapest edge out of the current tree.

Kruskal vs Prim: Comparison

Aspect	Kruskal's	Prim's
Strategy	Global edge sort	Local vertex expansion
Data structure	Union-Find (DSU)	Priority queue
Grows	A forest (multiple trees)	One tree from start node
Time	$O(E \log E)$	$O(E \log V)$ or better
Best for	Sparse graphs	Dense graphs (Fibonacci heap)
Matroid view	Direct graphic matroid greedy	Implicit; cut-property driven
Correctness basis	Matroid greedy theorem	Cut property

Both produce a minimum spanning tree. Kruskal aligns *directly* with the graphic matroid greedy algorithm; Prim's matroid interpretation is more implicit.

Part VII

Beyond MSTs

Matroid intersection · Submodularity · Generalisations

VII

Matroid Intersection

Definition: Matroid Intersection

Given two matroids $M_1 = (E, \mathcal{I}_1)$ and $M_2 = (E, \mathcal{I}_2)$ on the same ground set, find a maximum-weight set $S \in \mathcal{I}_1 \cap \mathcal{I}_2$.

Example: Bipartite Matching

$G = (U \cup V, E)$. A matching = edge set where each vertex appears ≤ 1 time.

M_1 : partition matroid on U — at most one edge per $u \in U$.

M_2 : partition matroid on V — at most one edge per $v \in V$.

Maximum matching = max common independent set of M_1 and M_2 .

Complexity:

- Single matroid: greedy, $O(|E| \log |E|)$.
- Matroid intersection: solvable in polytime via augmenting paths, $O(|E|^{1.5} \cdot T_{\text{oracle}})$.
- Three-matroid intersection: NP-hard in general!

Submodular Optimisation over Matroids

Definition: Submodular Function

$f : 2^E \rightarrow \mathbb{R}$ is **submodular** if for all $A, B \subseteq E$:

$$f(A \cup B) + f(A \cap B) \leq f(A) + f(B)$$

(equivalently: f has *diminishing marginal returns*).

Connection to matroids:

- The rank function r of any matroid is submodular.
- Maximising a monotone submodular function subject to a matroid constraint: greedy achieves a $(1 - \frac{1}{e}) \approx 0.632$ approximation!

Applications:

- **Influence maximisation** in social networks (viral marketing).
- **Feature selection** in machine learning.
- **Sensor placement** for maximum coverage.
- **Welfare maximisation** in combinatorial auctions.

Generalisations: Greedoids and Antimatroids

What if we relax matroid axioms?

Definition: Greedoid

$(E, \mathcal{F})$ with $\emptyset \in \mathcal{F}$ and: for $A, B \in \mathcal{F}$ with $|A| > |B|$, $\exists e \in A$ s.t. $B \cup \{e\} \in \mathcal{F}$.

Heredity (I2) is **not required**.

Examples of greedoids:

- Branching greedoids (rooted spanning trees explored in BFS order).
- Ear decompositions of graphs.
- Gaussian elimination sequences.

Definition: Antimatroid

Satisfies heredity (I2) but not exchange. Models “reachable” sets: start from a feasible set and expand.

Examples: convex point sets, node-search in trees.

Greedoids and antimatroids extend matroid theory to capture even more greedy-like algorithms beyond the classical matroid framework.

Applications of Matroid Theory

Application Domain	Matroid concept used
Network design (cable, roads)	Graphic matroid — MST
Bipartite matching	Matroid intersection
Job scheduling	Partition / transversal matroids
Error-correcting codes	Linear matroids over finite fields
VLSI circuit layout	Linear matroids, connectivity
Influence / viral marketing	Submodular optimisation over matroids
Approximation algorithms (TSP)	MST as 2-approximation
Phylogenetics	MST for tree reconstruction
Image segmentation	Graph-cut / matroid methods
Combinatorial auctions	Submodular welfare maximisation

Part VIII

Summary & Takeaways

The big picture

VIII

Key Takeaways

- A **matroid** $M = (E, \mathcal{I})$ abstracts the notion of independence from linear algebra, graph theory, and combinatorics.
- The three axioms — non-emptiness, heredity, and **exchange** — precisely characterise when greedy is globally optimal.
- All bases of a matroid have equal size, giving a consistent notion of **rank**.
- **Matroid Greedy Theorem (Edmonds 1971)**: Greedy finds a maximum-weight basis for every weight function iff the feasible sets form a matroid.
- **Graphic matroids** explain why Kruskal's and Prim's MST algorithms are correct.
- Matroids set the **exact boundary** between greedy algorithms that are provably correct and those that are merely plausible-looking.

References

Reference	Focus
Oxley, J. (2011). <i>Matroid Theory</i> , 2nd ed. Oxford University Press.	Comprehensive reference on matroid theory
Edmonds, J. (1971). “Matroids and the Greedy Algorithm.” <i>Math. Programming</i> 1, 127–136.	Original paper — matroid greedy theorem
Cormen, Leiserson, Rivest, Stein. <i>Introduction to Algorithms</i> , 4th ed. MIT Press.	MST algorithms, greedy, Union-Find
Schrijver, A. (2003). <i>Combinatorial Optimization</i> . Springer.	Matroids, polymatroids, submodularity
Korte, B. & Vygen, J. (2012). <i>Combinatorial Optimization</i> , 5th ed. Springer.	Greedoids, matroids, algorithms
Welsh, D. (1976). <i>Matroid Theory</i> . Academic Press.	Classic textbook — readable and complete
Lawler, E. (1975). “Matroid Intersection Algorithms.” <i>Math. Programming</i> 9, 31–56.	Matroid intersection algorithms

Appendix: Equivalent Axiom Systems

A matroid can be specified through several equivalent “lenses”. Each lens emphasizes a different kind of structure.

Independent sets $\mathcal{I}$

Main idea: feasible sets closed under taking subsets.

Exchange: a smaller independent set can be augmented from a larger one.

Bases $\mathcal{B}$

Main idea: maximal feasible sets all have the same size.

Basis exchange: swapping one element from one basis with a suitable element from another gives a basis again.

Circuits $\mathcal{C}$

Main idea: minimal dependent sets.

Elimination: two circuits sharing an element imply another circuit after eliminating that element.

Rank r

Main idea: dimension-like size of the largest independent subset.

Submodularity:

$$r(A \cup B) + r(A \cap B) \leq r(A) + r(B)$$

Closure cl

Main idea: span-like operator: which elements are forced by a set?

Mac Lane–Steinitz exchange:

$$e \notin \text{cl}(A), e \in \text{cl}(A \cup \{f\}) \Rightarrow f \in \text{cl}(A \cup \{e\})$$

- Different axiom systems are useful in different settings: circuits for graphic matroids, rank for linear algebra, bases for optimisation.
- The equivalences are constructive: each system can be converted into the others.